

MIGHTY LUBE

D-SERIES

POWER CONTROL SYSTEM

G12

INSTALLATION & MAINTENANCE MANUAL

V1.24 / 2.25



MIGHTY LUBE®
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WARNINGS

The following warnings are for the setup, use, grounding, maintenance, and repair of this equipment. The exclamation point symbol alerts you to a general warning and the hazard symbol refers to procedure-specific risk. Refer back to these warnings. Additional product specific warnings may be found throughout the body of this manual where applicable.

WARNING	
	EQUIPMENT MISUSE HAZARD Misuse can cause death or serious injury. • Do not operate the unit when fatigued or under the influence of drugs or alcohol. • Do not exceed the maximum working pressure or temperature rating of the lowest rated system component. See Technical Data in all equipment manuals. • Use fluids and solvents that are compatible with equipment wetted parts. See Technical Data in all equipment manuals. Read fluid and solvent manufacturer's warnings. For complete information about your material, request MSDS forms from distributor or retailer. • Check equipment daily. Repair or replace worn or damaged parts immediately with genuine manufacturer's replacement parts only. • Do not alter or modify equipment. • Use equipment only for its intended purpose. Call your distributor for information. • Route hoses and cables away from traffic areas, sharp edges, moving parts, and hot surfaces. • Do not kink or over bend hoses or use hoses to pull equipment. • Keep children and animals away from work area. • Comply with all applicable safety regulations.
	ELECTRIC SHOCK Improper grounding, setup, or usage of the system can cause electric shock. • Turn off and disconnect power at main switch before disconnecting any cables and before servicing equipment. • Connect only to grounded power source. • All electrical wiring must be done by a qualified electrician and comply with all local codes and regulations.
	ENTANGLEMENT HAZARD Rotating parts can cause serious injury. • Keep clear of moving parts. • Do not operate equipment with protective guards or covers removed. • Do not wear loose clothing, jewelry or long hair while operating equipment. • Equipment can start without warning. Before checking, moving, or servicing equipment, disconnect all power sources.

Introduction



The Mighty Lube® D-Series Power Control System is designed to supply low voltage power and system communication for connected lubrication / monitoring equipment. The D-Series may be supplied 120VAC* or 230VAC operation power.

FEATURES INCLUDE:

- 7" touch screen user interface
- Visual level indicator
- Audible fault indicator
- Available PLC outputs
- LAN network connection
- Contained in powder coated steel enclosure

**Standard 120VAC units include a 6 foot US style cord.*



INSTALLATION

LOCATION

The proper location for the Mighty Lube® Power Control System is free from moisture, excessive heat or vibration. The unit requires constant operational voltage (120VAC or 230VAC) and an **optional LAN network connection***. The location must be out of the main lift truck traffic area and accessible for maintenance. (Next to a building column or in a corner is recommended.) Lubricator/Monitor distance not to exceed 300 feet from the power supply (wire distance).

CONNECTION *(See Wiring Diagram on page 17)*

The D-Series Power Control System may be supplied with 120VAC or 230VAC single phase operational voltage. This system may be hard wired if desired. The Lubricator/Monitor is connected via 4-conductor cable. Connected devices must be connected in a “daisy-chain” configuration or from the first to the second and so on. A maximum of 8 lubrication/monitor heads may be connected, not to exceed a 300 foot radius.



START-UP

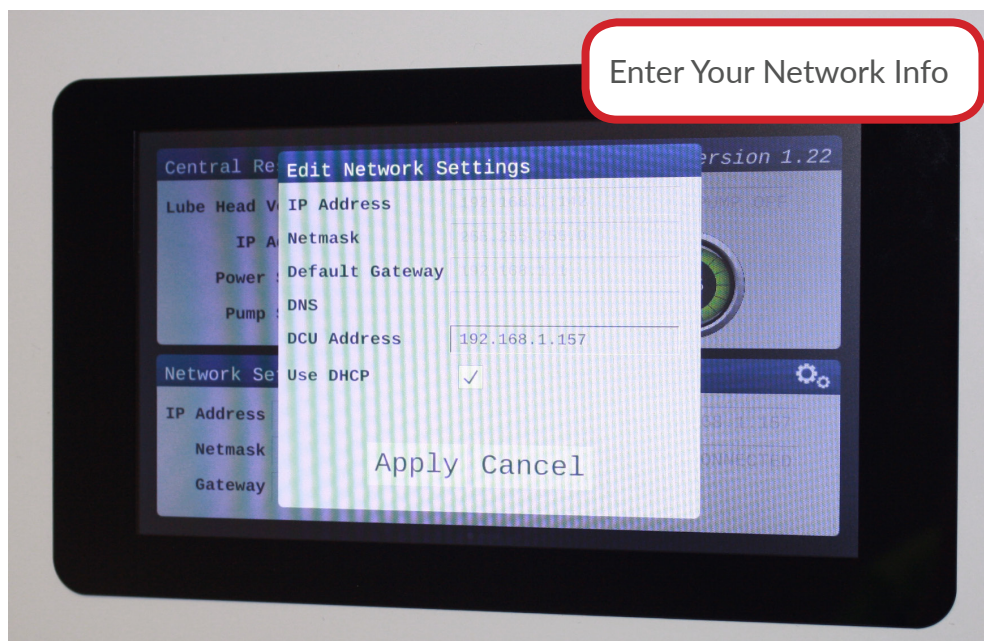
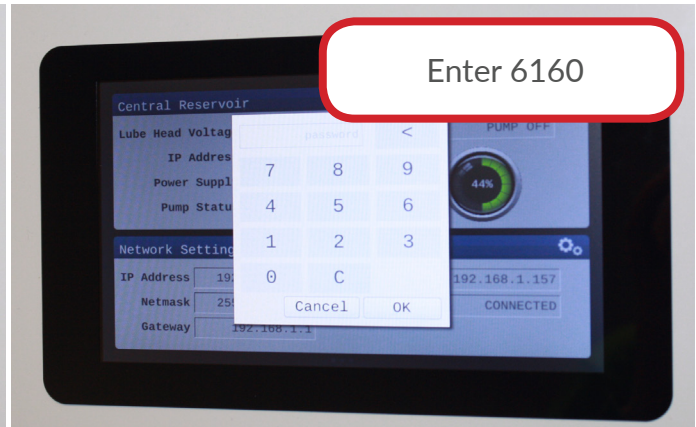
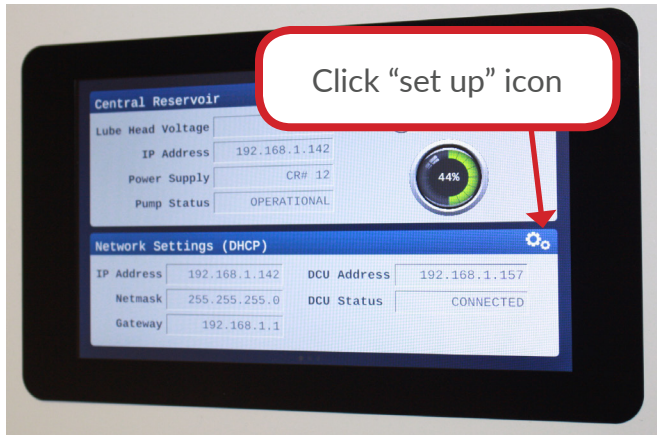
Power On

System startup will be signaled by one short audible tone. The low voltage and optional lube pump motor circuits will be energized if applicable. The LAN connections will be configured as programmed. The MAIN screen will display program version, system controller voltage, available reservoir level and network configuration. Full system function may require 2 minutes from start-up, depending upon system configuration.

****Some features require a network connection.***



SET-UP



Network

The Mighty Lube D-Series Power Control System allows the user to configure LAN network from the front mounted touchscreen. Press the “setup” icon located to the right-middle of the main screen. Enter password 6160 at prompt, press enter. Enter all network specifics required for operation on your company network. Press “apply” to exit Network Setup screen and implement changes to network configuration.



SET-UP



Lubrication/Controller Interface

Depending on system configuration, the touch-screen may be used to adjust all connected lubrication controller settings including optional local lubrication. The D-Series Power Control System may be used as a single lubricator, a central reservoir/power supply or a combination of both.

From the MAIN screen, press down and slide to the left side of the screen to view the lube level screen. Repeating this action will move to the controller configuration screen. Use the left/right arrows, at the top of the screen, to select the local (power supply) or remote (any connected 9100(+) series) settings. Conveyor data is displayed on the left of the screen (length, pitch, speed, drive). Menu ↑↓ arrows may be used to move through the lubrication menu. Value ↑↓ keys will adjust settings.



SET-UP

Lubrication Menu

LUBRICATION MENU	DESCRIPTION
N1 - INPUT 1 (LINK NUMBER)	N1 number refers to the number of chain link sets to be counted in one complete revolution of the conveyor. This value is automatically entered into the program if the EOC (END OF CHAIN) magnetic block is properly installed on the chain with the magnet facing the switch and that the switch is properly adjusted to sense the block as it passes. The EOC function must be set to Y in the setup menu.
C1 - OUTPUT 1 (OFF-CYCLES)	Off-cycles for input 1 refers to the number of revolutions the conveyor will make between applications of lubricant. Scroll to C1 by pushing the menu down arrow. Using the value up/down switch, program the number of desired link off-cycles. For constant lubrication, enter 0000 or 0001. To lubricate every other revolution of the chain, enter 0002. For every fifth revolution, enter 0005 and so on. The number of cycles without lubrication plus one on-cycle.
T1 - INPUT 1 (LUBE TIME)	Lube time refers to how long the solenoid valve sprays lubricant onto the chain as it passes by the lubricator. Scroll to T1 by pushing the menu down arrow. Use the value up/down switch to program the duration in milliseconds, of the solenoid operation for each link that is being lubricated. Typical settings are about 0010 to 0015 for I-beam systems. It is recommended that this time be set no lower than 0008 to insure proper activation of the solenoid.
D1 - LINK SWITCH DELAY TIME	Delay time refers to how long the switch must wait before counting again. This feature is designed to prevent the valve from "double firing" and dispensing lubricant to an undesirable location. Scroll to D1 by pushing the menu down arrow. Use the value up/down switch to program the delay time for the link switch. However, if the time is too long, you may miss the next link. Link delay time is adjustable from 0.000 seconds (no delay) to 3.999 seconds (maximum delay).



SET-UP

LUBRICATION MENU	DESCRIPTION
N2 – INPUT 2 (TROLLEY OR BEARING NUMBER)	N2 number refers to the exact number of trolley bearing pairs/sets (I-beam conveyor) in one complete revolution of the conveyor. Scroll to N2 by pushing the menu down arrow. Using the value up/down switch, program the number of trolleys on the chain. If the N2 value is set at 0000, this cycle will automatically be updated along with the N1 (links). If the exact number is not known, see the following ALTERNATE METHOD OF PROGRAMMING N2 (TROLLEY NUMBERS).
ALTERNATE METHOD OF PROGRAMMING N2 (TROLLEY NUMBERS)	1. Program 9999 into N2 (trolley number) portion of the control board using the value switches. 2. Mark or place a tag on the conveyor directly in front of the lubricator. This allows you to keep track of the conveyor's progress. 3. Press the reset switch to bring counters to zero. 4. Start the conveyor and let it run one complete revolution until the mark or tag appears in the same place in front of the lubricator. Note the "C" (or Count) number in the display. This is the exact number to be programmed into the N2 (trolley number) portion of the control board. Use the value up/down switch to enter the desired number. 5. Programming N2 (trolley number) is now complete.
C2 – OUTPUT 2 OFF-CYCLES	Off-cycles refer to the number of revolutions the conveyor will make between applications of lubricant. Scroll to C2 by pushing the menu down arrow. Using the value up/down switch, program the number of desired trolley off-cycles. For constant lubrication, enter 0000 or 0001. To lubricate every other revolution of the chain, enter 0002. For every fifth revolution, enter 0005 and so on. Normally programmed to same number as C1.



SET-UP

LUBRICATION MENU	DESCRIPTION
T2 - OUTPUT 2 LUBE TIME	Lube time refers to how long the solenoid valve sprays lubricant onto the trolley bearings as they pass the lubricator. Scroll to T2 by pushing the menu down arrow. Use the value up/down switch to program the duration, in milliseconds, of solenoid operation for each trolley or vertical bearing that is being lubricated. Typical settings are about 0010 0015 for I-beam systems. It is recommended that this time be set no lower than 0008 to insure activation of the solenoid.
D2- INPUT 2 DELAY TIME	Delay time refers to how long the switch must wait before counting again. Scroll to D2 by pushing the menu down arrow. Use the value up/down switch to program the delay time for the trolley switch. This function prevents the switch from reactivating for the programmed number of milliseconds. This is useful for preventing double firing resulting in lubricant being dispensed in an undesired location, or the same location from being over lubricated. D2 delay time is adjustable from 0.000 seconds (no delay) to 3.999 seconds (maximum delay).
NC - CAT DRIVE NUMBER	Cat drive number refers to the number of engagement dogs on the cat drive chain. Scroll to NC by pushing the menu down arrow. Using the value up/down switch, program the number of drive dogs. <i>Note: The main cat drive solenoid valve is activated by the lubricator link switch. It does not require its own switch to operate. On units with a single drive, the cat drive lubricator need only be wired in accordance with the schematic found in the control box. If main and back-up drives are used, the lubricator with the switch must be installed on the back-up drive and wired according to the schematic. No additional programming is required if the back-up drive is used.</i>
CC - CAT DRIVE OFF-CYCLES	Off-cycles refer to the number of revolutions the cat drive chain makes between applications of lubricant. Scroll to CC by pushing the menu down arrow. Using the value up/down switch, program the number of desired off cycles. For constant lubrication, enter 0000 or 0001. To lubricate every other revolution of the chain, enter 0002. For every fifth revolution, enter 0005 and so on. A usual setting for a drive is 1000 to 3000 depending on the environment.



SET-UP

LUBRICATION MENU	DESCRIPTION
TC - CAT DRIVE LUBE TIME	Lube time refers to how long the solenoid valve sprays lubricant onto the cat drive chain. Scroll to TC by pushing the menu down arrow on the menu screen. Use the value up/down switch to program the duration, in milliseconds, of solenoid operation for each dog and link that is being lubricated. It is recommended that this time be set no lower than 0008 to insure activation of the solenoid.
N3 - INPUT 3 AUXILIARY NUMBER	This function is useful for lubricating such things as free carriers or spinners and requires the use of an additional switch to sense the additional lubrication point(s). For this auxiliary example we will use a free carrier system. Auxiliary number refers to the number of free carriers to be counted in a power and free conveyor system. Scroll to N3 by pushing the menu down arrow. Using the value input switches, program the number of auxiliary bearings.
C3 - INPUT 3 OFF-CYCLES	Off-cycles refer to the number of times all of the free carrier will pass the lubricator between applications of lubricant. Scroll to C3 by pushing the menu down arrow on the menu screen. Using the value up/down switch, program the number of desired off cycles. For constant lubrication, input 0000 or 0001. To lubricate every other revolution of the chain or cycle of the free carriers, enter 0002. For every fifth revolution or cycle, enter 0005 and so on.
T3 - OUTPUT 3 LUBE TIME	Lube time refers to how long the auxiliary valve will spray lubricant onto the bearings and guide rollers of the free carrier as it passes the lubricator. Scroll to T3 by pushing the menu down arrow. Use the value up/down switch to program the duration, in milliseconds, of solenoid operation for each point that is being lubricated. It is recommended that this time be set no lower than 0008 to insure activation of the solenoid.



SET-UP

LUBRICATION MENU	DESCRIPTION
D3 – INPUT 3 SWITCH DELAY TIME	Delay time refers to how long the switch must wait before counting again. This function prevents the switch from reactivating for the programmed number of milliseconds. This is useful for preventing double firing resulting in lubricant being dispensed in an undesired location or the same location from being over lubricated. Scroll to D3 by pushing the menu down arrow. Use the value up/down switch to program the delay time for the link switch.
N4 – INPUT 4 AUXILIARY	This function is useful for lubricating such things as free carriers or spinners and requires the use of an additional switch to sense the additional lubrication point(s). For this auxiliary example we will use a free carrier system. Auxiliary number refers to the number of free carriers to be counted in a power and free conveyor system. Scroll to N4 by pushing the menu down arrow. Using the value input switches, program the number of auxiliary bearings.
C4 – INPUT 4 OFF-CYCLES	Off-cycles refer to the number of times all of the free carrier will pass the lubricator between applications of lubricant. Scroll to C4 by pushing the menu down arrow. Using the value up/down switch, program the number of desired off cycles. For constant lubrication, input 0000 or 0001. To lubricate every other revolution of the chain or cycle of the free carriers, enter 0002. For every fifth revolution or cycle, enter 0005 and so on.
T4 – OUTPUT 4 LUBE TIME	Lube time refers to how long the auxiliary valve will spray lubricant onto the bearings and guide rollers of the free carrier as it passes the lubricator. Scroll to T4 by pushing the menu down arrow. Use the value up/down switch to program the duration, in milliseconds, of solenoid operation for each point that is being lubricated. It is recommended that this time be set no lower than 0008 to insure activation of the solenoid.
D4 – INPUT 4 DELAY TIME	Delay time refers to how long the switch must wait before counting again. This function prevents the switch from reactivating for the programmed number of milliseconds. This is useful for preventing double firing resulting in lubricant being dispensed in an undesired location or the same location from being over lubricated. Scroll to D4 by pushing the menu down arrow. Use the value up/down switch to program the delay time for the link switch.



SET-UP

LUBRICATION MENU	DESCRIPTION
Chain FT/M	000.0 - 299.9 Conveyor Speed limit is different for individual conveyors.
10' LN0000	Displays wear of the last 10' section of chain measured.
LBL LN0000	Displays wear of the last link measured.
DRIVE AMPS	Displays the real-time cat drive motor current if enabled.
AIR PSI	Displays the current air pressure readings on air take-up if enabled.
MAIN HRS	Number of hours that the main drive has been operating.
BKUP HRS	Number of hours that the backup drive has been operating.
COM ADDR	Displays station address and current communications status. Each monitor Head must have a different station address number. The station address can be changed in SETUP menu.



Set-Up Menu

Setup key opens the initial system setup menu. Use value ↑ ↓ keys to enter password 6160, press setup again to submit password. Use menu ↑ ↓ keys to edit setup options. Press setup again to exit and save changes to setup program.

INITIAL SET-UP			
SET-UP MENU	MENU VALUES	FACTORY SETTING	DESCRIPTION
LUBE MENU	Y or N	Y	Turns on lubrication portion of system.
10' WEAR*	Y or N	N	Activates 10' wear monitor.
LBL WEAR*	Y or N	Y	Activates LBL wear monitor.
WEAR SPD MIN	Numerical	05	Minimum Speed of chain wear monitoring.
SPD TIME OUT	Numerical	30	Delay before alarm trigger after start-up.
MONITOR AMPS*	Y or N	Y	Activates drive amp monitor.
MONITOR AIR*	Y or N	Y	Activates take-up air pressure monitor.
MONITOR DIST*	Y or N	Y	Activates take-up distance monitoring.
ENABLE N2 ERR			Trolley Error input indicator
ENABLE EOC SW	Y or N	Y	Allows EOC counter reset.
10' BEFORE N1*	Y or N	N	Switches 10' wear switch placement to after (according to direction of travel) N1 switch.
PAINT MENU	Y or N	N	Paint Chain Menu

* Some SET-UP options may not be available for all Lubricators/Monitors.



SET-UP

SET-UP MENU	MENU VALUES	FACTORY SETTING	DESCRIPTION
STATION ADDR	1-99		EACH CENTRAL SYSTEM MONITOR MUST HAVE A DIFFERENT STATION ADDRESS.
CHAIN PITCH	2.1 - 99.1		Length of distance from the start to finish of chain link. SET AS NEEDED.
LBL ADJUST	Numerical	0.00	Add or remove up to 0.25" to each chain link.
DIR LUBE TM	1-999	60 seconds	Length of time solenoid valves will operate during test function.
USE AMPS C1*	Y or N	N	Allows lubrication cycle control by drive motor current.
USE AMPS C2*	Y or N	N	Allows lubrication cycle control by drive motor current.
EXT LUBE N1	Y or N		External Lube Control N1
EXT LUBE N2	Y or N		External Lube Control N2 (Trolley Greaser)
EXT LUBE N3	Y or N		External Lube Control N3 (Carrier Greaser)
EXT LUBE N4	Y or N		External Lube Control N4 (Guide Greaser)
SPEED DELAY	Y or N	N	Allows for lubrication offset as conveyor speed changes.
ENABLE P CHG	Y or N	Y	Allows user to change P values.
SAVE P VALUES*	Y or N	Y	Saves P values as a starting point.

* Some SET-UP options may not be available for all Lubricators/Monitors.



SET-UP

SET-UP MENU	MENU VALUES	FACTORY SETTING	DESCRIPTION
RESTORE P VAL*	Y or N	Y	Changes P values to saved settings.
SWITCH STATUS #1		N1 - - - - -	Displays which switches are activated. Switch name/number will be displayed on screen when active.
SWITCH STATUS #2		- - LBL - - -	Displays which switches are activated. Switch name/number will be displayed on screen when active.
ANALOG INPUT #1		Distance, Air, Amps	Analog inputs are variable and may be configured for different devices.
ANALOG INPUT #2		Distance, Air, Amps	Analog inputs are variable and may be configured for different devices.
ANALOG INPUT #3		Distance, Air, Amps	Analog inputs are variable and may be configured for different devices.
OUTPUT TEST <i>(toggle selection ON by pressing Direct Lube and change selected output by pressing value keys)</i>	ON/OFF	N1: OFF S2: OFF S1: OFF CATB: OFF CATM: OFF N4: OFF N3: OFF N2: OFF	The output test confirms operation of the solenoid valves without interrupting the program.
LUBE VOLTAGE		+/- 24.00V	Displays available low voltage power.
TANK LEVEL		0-100% 0-2V	Displays level sensor input voltage and corresponding level percentage.

* Some SET-UP options may not be available for all Lubricators/Monitors.

PRESS **SETUP** AGAIN TO EXIT SETUP MENU AND SAVE CHANGES.



SET-UP

DIRECT LUBE key will begin “test” lube cycle for present line based on lubrication settings. Lubrication system direct lube function requires active switch inputs and proper system setup.

The **CHAIN DATA** key will display the individual conveyors chain pitch and real time chain speed.

The lubricator program **RESET** key will reset all counts to 0000 and begin a lubrication cycle of that conveyor. This reset will not effect N1 counts if End of Chain Switch is enabled.



PROGRAM VERSION

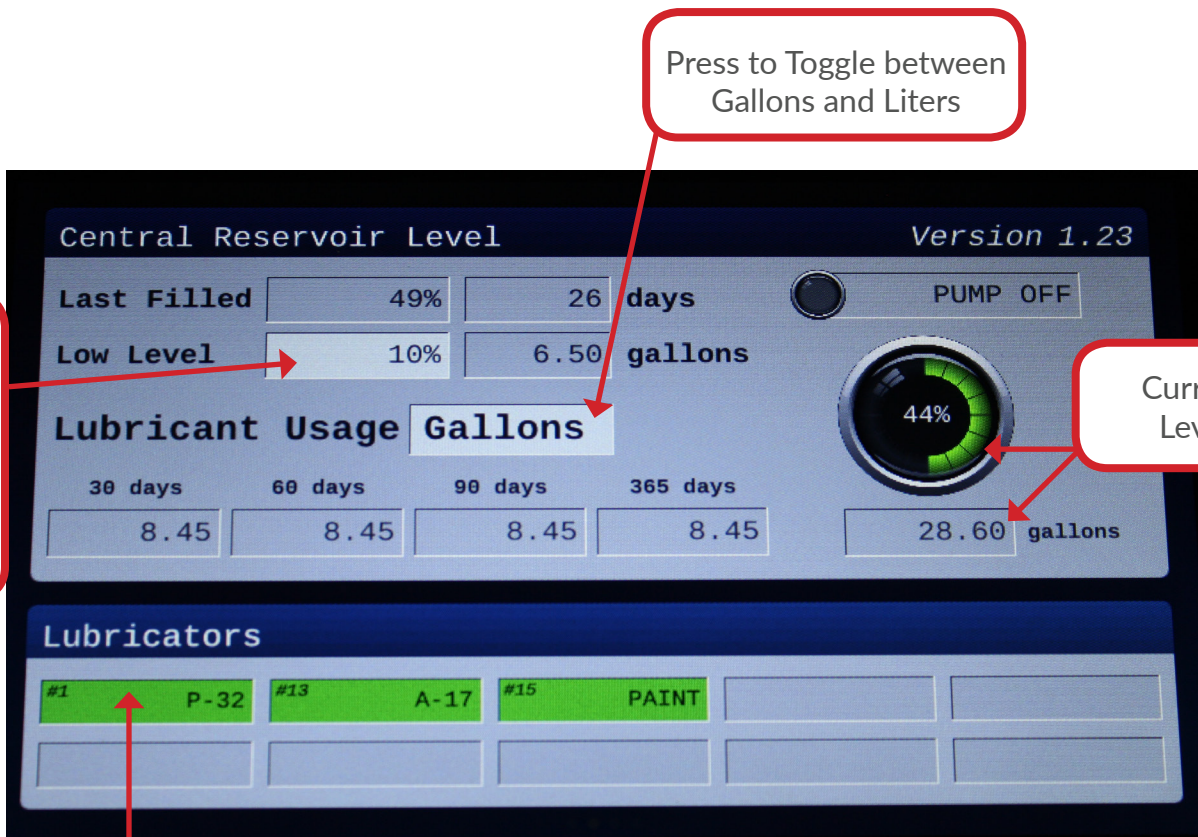


The D-Series Power Control System program version can be verified at the PROGRAM IDENTIFICATION screen. To view the program versions currently running on the Mighty Lube® D-Series Power Control System, press the touch screen and slide your finger to the left three times from the MAIN screen. This screen also displays contact information for Mighty Lube Systematic Lubrication, Inc. and system time and date.



LEVEL HISTORY

The Mighty Lube® D-Series Power Control System will track lubricant usage up to one year and display this information on the lube level screen. The reservoirs current fill level will be displayed to the right as a percentage and in volume (gallons/liters). The lubricant history is displayed, including percentage and elapsed time since the reservoir was last filled. The total reservoir consumption spanning the previous 30, 60, 90 and 365 days is displayed. Connected lubrication units are displayed at the bottom of the screen with associated station address. Touch green icons in the Lubricators section to name the head units connected to conveyors. Any unnamed station address will disappear if communication is lost.



Specific system functions will vary with purchased options.



LEVEL HISTORY

Press Here to Enter Low Level Alarm Value

Low Level Alarm Settings
Press to Alter Low Level Set Point

Edit Low Lube Level Alarm

New Alarm Value

Apply

Cancel

Press Apply to Complete Changes

Central Reservoir Level

Last Filled 49% 26 days PUMP OFF
Low Level 10% 6.50 gallons
Lubricant Usage Gallons
30 days 60 days 90 days 365 days
8.45 8.45 8.45 8.45 28.60 gallons

Lubricators

#1 P-32 #13 A-17 #15 PAINT

Active Monitor Head Unit

Press to Name Connected Head Units

Edit Lubricator Name

Name

1 2 3 4 5 6 7 8 9 0
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M <
space - _ # () | []

Apply Cancel



1. SET PSI TO OPERATION PSI— 10%.
2. SET DIP 1 TO ON FOR PUMP FUNCTION.
3. DIP 2 TO ON FOR 65 GALLON RESERVOIR.



MIGHTY LUBER®

SYSTEMATIC LUBRICATION

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ALL INFORMATION CONTAINED HEREIN IS UNCLASSIFIED DATE 05-01-2008 BY 60322 UCBAW

MTYL _____

SCALE _____

TRAVERSE POINT _____

EXAMPLE 3.00/5 _____

NO. OF BALLS _____

NO. OF BALLS IN CONTACT _____

NO. OF BALLS IN CONTACT $\times$ 5 _____

NO. OF BALLS IN CONTACT $\times$ 10 _____

NO. OF BALLS IN CONTACT $\times$ 20 _____

NO. OF BALLS IN CONTACT $\times$ 40 _____

NO. OF BALLS IN CONTACT $\times$ 80 _____

NO. OF BALLS IN CONTACT $\times$ 160 _____

NO. OF BALLS IN CONTACT $\times$ 320 _____

NO. OF BALLS IN CONTACT $\times$ 640 _____

NO. OF BALLS IN CONTACT $\times$ 1280 _____

NO. OF BALLS IN CONTACT $\times$ 2560 _____

NO. OF BALLS IN CONTACT $\times$ 5120 _____

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NO. OF BALLS IN CONTACT $\times$ 193428131138340667952988160 _____

NO. OF BALLS IN CONTACT $\times$ 386856262276681335905976320 _____

NO. OF BALLS IN CONTACT $\times$ 773712524553362671811952640 _____

NO. OF BALLS IN CONTACT $\times$ 1547425049106725343623905280 _____

NO. OF BALLS IN CONTACT $\times$ 3094850098213450687247810560 _____

NO. OF BALLS IN CONTACT $\times$ 6189700196426901374495621120 _____

NO. OF BALLS IN CONTACT $\times$ 12379400392853802748991242240 _____

NO. OF BALLS IN CONTACT $\times$ 24758800785707605497982484480 _____

NO. OF BALLS IN CONTACT $\times$ 49517601571415210995964888960 _____

NO. OF BALLS IN CONTACT $\times$ 99035203142830421991939777920 _____

NO. OF BALLS IN CONTACT $\times$ 198070406285660843983879555840 _____

NO. OF BALLS IN CONTACT $\times$ 396140812571321687967759111680 _____

NO. OF BALLS IN CONTACT $\times$ 792281625142643375935518223360 _____

NO. OF BALLS IN CONTACT $\times$ 158456325



TROUBLESHOOTING

PROBLEM	SOLUTION(S)
No Function / No Power	• Check Incoming AC Voltage - Confirm 120 or 230VAC • Check 3AMP Fuse • Confirm On/Off Switch Function/Position • Replace PCS-8D
No Com to Pump Board	• Check UART Connection - Cycle Power (15+ Seconds Off) • Replace PCS-8D
Blank Touchscreen after 3 Min	• Confirm LED indicator activity • Check 24 VDC Input to Pump Board • Check 5 VDC Output to Display • Check UART Connection - Cycle Power (15+ Seconds Off) • Replace Touchscreen RPI Assembly
No Comm to Head Boards	• Confirm 4 Conductor Cable Connection (See Wiring Page 18) • Confirm 24 VDC Power to Head Board (Check Fuse) • Check Head Board Station Address - 1 to 99 • If no comm to all boards: Replace PCS-8D • If no comm to single board: Replace head board
No Pump Function (No Alarm)	• Confirm VAC output from PCS-8D, pump / new contacts • Check 10 AMP Fuse • Check Head/Lubricator Program for Lubrication Cycle • Confirm Conveyor Chain Speed (<i>Pump will shut down when chain is stopped</i>) • Confirm Dip #1 to on Position • Confirm pump on indicator at touchscreen home page • Replace PCS-8D

Troubleshooting Continued on Next Page

***Contact Mighty Lube Systematic Lubrication or Authorized Distributor for Service.**



TROUBLESHOOTING

PROBLEM	SOLUTION(S)
No Pump Function (Alarm)	» NO PRESSURE • Check Reservoir Fill Level • Check Lube Line Integrity • Check Lube Line Regular/Pressure Switch Setpoint/Function » NO COM • Check Head/Lubricator Connection • Check 24 VDC Power • Confirm Unique Station Address at Each Head/Lubricator • Replace PCS-8D
Reservoir Level 0% on Screen	• Confirm Dip #1 Positioned to the Right/On • Reservoir Empty / 3" or less • Check Pressure Transducer Connection at A4 (See Wiring Page 18) • Confirm Pressure Transducer Function, Measure 22-24VDC Between BN and BL • Check Level Tube/Fittings for Leaks • Replace Fittings/Sealant <i>Reset level pressure transducer after repairs by lifting pump assembly from reservoir to clear lubricant obstruction.</i> • If transducer voltage is measured below incoming voltage: Replace PCS-8D • If transducer voltage is measured the same as incoming voltage: Replace Transducer
Reservoir Level Incorrect	• Check level tube / fittings for leaks • Confirm Dip #2 Position • Left/Off 10 Gallon Reservoir • Right/On 65 Gallon Reservoir • Reset level pressure transducer by lifting pump assembly from reservoir and allow fluid to drain, retest function.

****Contact Mighty Lube Systematic Lubrication or Authorized Distributor for Service.***



TROUBLESHOOTING

PROBLEM	SOLUTION(S)
Unable to Save Network Options	Missing or defective USB memory backup • Confirm USB memory stick is installed in one of 4 USB ports • Replace USB memory backup with new USB drive (8GB or larger)
No Reservoir Level History	Missing or defective USB memory backup • Confirm USB memory stick is installed in one of 4 USB ports • Replace USB memory backup with new USB drive (8GB or larger)
Head Unit Name Lost at Power Cycle	Missing or defective USB memory backup • Confirm USB memory stick is installed in one of 4 USB ports • Replace USB memory backup with new USB drive (8GB or larger)

QUESTIONS?

**CONTACT
MIGHTY LUBE**



MIGHTY LUBE®

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